



BACHELOR THESIS TOPIC

Student: **Ostap Pelekh**
Student's ID: 128102
Study programme: Applied Informatics
Study field: Computer Science
Thesis supervisor: Ing. Roderik Ploszek, PhD.
Head of department: doc. Ing. Milan Vojvoda, PhD.
Workplace: Institute of Computer Science and Mathematics

Topic: **Local LLM-based Terminal Assistant**

Language of thesis: English

Specification of Assignment:

The aim of this thesis is to test the use of local LLMs in the context of a terminal assistant. As part of this thesis, it is necessary to design and implement a CLI/TUI assistant based on a local LLM, the purpose of which will be to execute user commands in natural language, obtain relevant information, and generate commands for the Linux OS terminal. Users will be able to use the assistant to streamline their work in the terminal through intelligent command suggestions and obtain answers to questions related primarily to the terminal.

Tasks:

1. Research and analyze existing terminal assistant solutions using LLM, identify their strengths and weaknesses. Explore the possibilities and limitations of local LLMs.
2. Design the architecture of the terminal assistant, including integration with the selected local LLM.
3. Implement the CLI/TUI terminal assistant according to the proposed architecture.
4. Evaluate the effectiveness of the proposed solution.

Selected bibliography:

1. NAVEED, Humza; KHAN, Asad Ullah; QIU, Shi; SAQIB, Muhammad; ANWAR, Saeed; USMAN, Muhammad; AKHTAR, Naveed; BARNES, Nick; MIAN, Ajmal. A Comprehensive Overview of Large Language Models [online]. 2024. Version 10 [visited on 2026-02-03]. Available from doi: 10.48550/arXiv.2307.06435.
2. ZHAO, Wayne Xin; ZHOU, Kun; LI, Junyi; TANG, Tianyi; WANG, Xiaolei; HOU, Yupeng; MIN, Yingqian; ZHANG, Beichen; ZHANG, Junjie; DONG, Zican; DU, Yifan; YANG, Chen; CHEN, Yushuo; CHEN, Zhipeng; JIANG, Jinhao; REN, Ruiyang; LI, Yifan; TANG, Xinyu; LIU, Zikang; LIU, Peiyu; NIE, Jian-Yun; WEN, Ji-Rong. A Survey of Large Language Models [online]. 2025. Version 16 [visited on 2026-02-03]. Available from doi: 10.48550/arXiv.2303.18223.

Deadline for submission of Bachelor thesis:	05. 06. 2026
Approval of assignment of Bachelor thesis:	28. 04. 2026
Assignment of Bachelor thesis approved by:	doc. Mgr. Daniela Chudá, PhD. – Study programme supervisor